

Day 07 of 60

80% of ML is data prep. Here's what that really looks like.

The model is only as good as the data behind it. This carousel shows the lifecycle, the definition of clean data, a real before/after example, recruiter expectations, and exactly what to showcase.

 5 slides

 4 Min Read

 Career-ready

Lifecycle map

See the step-by-step flow from raw source to model-ready dataset.

Clean data meaning

Learn the four qualities that make data useful for ML.

Recruiter radar

What hiring managers are actually looking for in data-focused roles.



From raw source to model-ready input

RAW SOURCE

Logs, spreadsheets, APIs, sensors — data arrives messy, incomplete, and inconsistent.

INGEST + VALIDATION

Collect, standardize, and verify values before anything else can be trusted.

CLEAN + ENRICH

Fix duplicates, normalize formats, fill gaps, and derive features that matter.

FEATURE-READY

Transform columns into model inputs and make sure every row is complete and consistent.

Why this matters

A model trained on garbage data learns garbage patterns. Clean data is the foundation of accuracy, explainability, and production stability.

Clean data is not just “no missing values”

- 1 **Accurate:** values reflect reality, not noise or mislabeled entries.
- 2 **Consistent:** formats, units, and categories follow the same rules across every row.
- 3 **Complete:** required fields are filled, or missingness is intentionally handled.
- 4 **Feature-ready:** variables are engineered, encoded, and scaled for the model you will use.

The full definition

Clean data is trustworthy, structured, and purpose-built for the ML problem — not just pretty-looking rows in a spreadsheet.

Turn noisy source data into a model-ready table

Before

Date values in mixed formats, missing city names, inconsistent category labels, and duplicate rows from repeated uploads.

After

Standardized dates, filled missing values, normalized categories, deduplicated rows, and a clean target variable ready for training.

What changed

This is the real work: removing ambiguity, enforcing structure, and making sure the algorithm sees consistent signals instead of random noise.

The data prep strengths that stand out

WHAT RECRUITERS WANT

SQL + ETL pipelines, data cleaning tools, feature engineering, validation checks, and experience with messy real-world datasets.

WHAT OFTEN GETS MISSED

Data contracts, schema drift handling, reproducible preprocessing, error logging, and annotation of assumptions.

The gap

Many candidates talk about models. Few show they can make data reliable in production — and that difference is what hiring teams notice.

Your data prep portfolio should prove process, not just output

- 1** Pipeline walkthrough: share the steps, tools, and decisions from raw source to training dataset.
- 2** Before/after samples: show the messy input and the clean table the model actually used.
- 3** Validation checks: demonstrate how you caught bad values, duplicates, and drift before training began.
- 4** Business impact: explain why the cleanup mattered for accuracy, reliability, or onboarding a new model.

Final note

If your next interview asks "How do you prepare data?", answer with a concrete pipeline, a cleanup example, and a clear business result.